

Jack Foreback

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 [Jack Foreback](#) |  [JackForeback](#)

Allendale, Michigan - 49401, USA

EDUCATION

• BS Computer Science and Mathematics

August 2023 - May 2027

Grand Valley State University

Allendale, MI

◦ GPA: 4.0

◦ Relevant Coursework: Numerical Analysis, Graph Theory, Data Structures and Algorithms, System-Level Programming and Utilities, Analysis of Differential Equations, Algorithm Engineering

RESEARCH EXPERIENCE

• REU on NeuroAI

June, 2026 - August, 2026

Rochester Institute of Technology



- Developed a Recurrent Spiking Predictive Coding model and training software in Jax.
- Gathered training data and model statistics demonstrating the necessity of recurrence in spiking predictive coding.

• REU on Bias in Classification

June, 2025 - August, 2025

Brown University



- Investigated the role of bias in Per-Class Classification accuracy and Max-Min accuracy.
- Developed methods to mitigate and understand these model biases.

• Undergraduate Research on AI in Bioinformatics

November, 2023 - December, 2024

Grand Valley State University



- Tested Attention and Transformers in a Multi Modal Single Cell Variational Autoencoder(VAE).
- Created a baseline Mixture of Experts Variational Autoencoder.

OTHER WORK EXPERIENCE

• Summer Tech Department

June 2023 – August 2023

Goodrich High School

- Set up PCs and ensured their proper functioning
- Replaced connections for the district's internet servers

PRESENTATIONS

• Recurrent Spiking Predictive Coding for Sparse Representations

July 30th, 2026

RIT Undergraduate Research Symposium



• Linear Classification Can Lack Fairness Even with Linearly Separable Data

August 1st, 2025

Brown Summer Research Symposium



• Linear Classification Can Lack Fairness Even with Linearly Separable Data

July 24th, 2025

Leadership Alliance National Symposium



• Testing Combinations of Transformers and Deep Learning in Bioinformatics

Jan 8-11, 2025

Joint Mathematics Meetings



• A Cross-Species Foundation Model for Single-cell Transcriptomics

November 2nd, 2024

West Michigan Regional Undergraduate Science Research Conference



• Multi-Head Attention in Single Cell Variational Autoencoders

July 24-26, 2024

Bioconductor North America



• Multi-Head Attention in Single Cell Variational Autoencoders

July 26, 2024

Grand Valley State University Summer Scholars Showcase



ACTIVITIES

- **Lower Michigan Mathematics Competition** *April, 2026*
- **Training AI Agents To Play Minecraft** *January, 2026*
- **ICPC East Central North America Regional Contest** *November, 2025*
- **Evolving Bitstrings** *January, 2025*
- **ICPC East Central North America Regional Contest** *November, 2024*
- **Coding Backpropagation By Hand In C** *January, 2024*

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C, Bash, Git, AWK, JavaScript, R
- **Libraries:** Pytorch, Pytorch Lightning, Jax
- **Job Related Skills:** Public Speaking, Creative and Technical Writing, Spanish

HONORS AND AWARDS

- **Presidential Scholarship**, Grand Valley State University *May, 2023-Present*
- **Dean's List**, Grand Valley State University *May, 2023-Present*